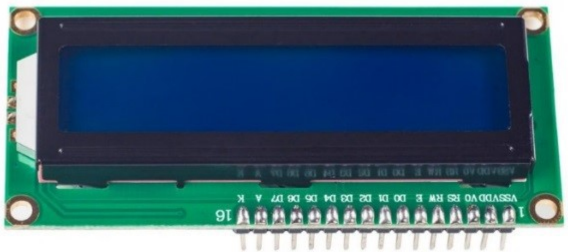
LCD1602 Module

From Wiki

Contents

Introduction



LCD1602, or 1602 character-type liquid crystal display, is a kind of dot matrix module to show letters, numbers, and characters and so on. It's composed of 5x7 or 5x11 dot matrix positions; each position can display one character. There's a dot pitch between two characters and a space between lines, thus separating characters and lines. The model 1602 means it displays 2 lines of 16 characters.
Generally, LCD1602 has parallel ports, that is, it would control several pins at the same time. LCD1602 can be categorized into eight-port and four-port connections. If the eight-port connection is used, then all the digital ports of the SunFounder Uno board are almost completely occupied. If you want to connect more sensors, there will be no ports available. Therefore, the four-port connection is used here for better application.

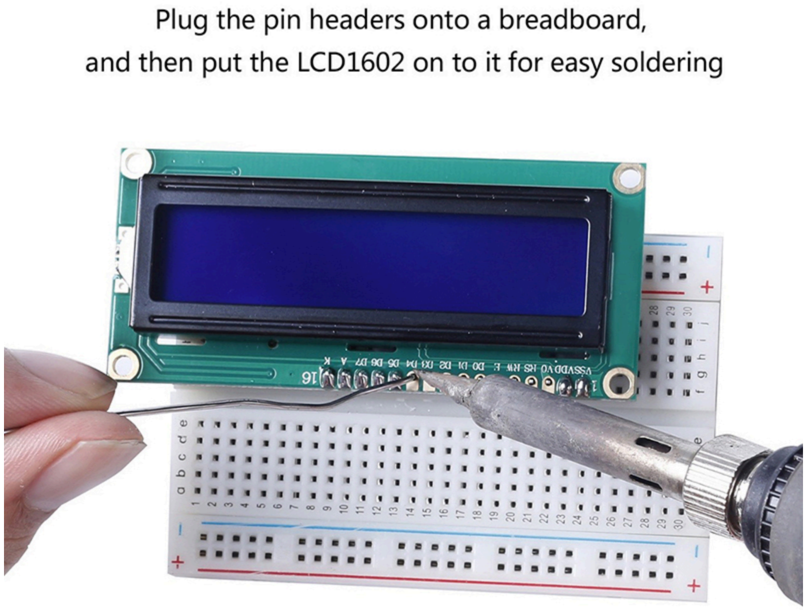
Pins Functions

Pin	Function
VSS	connected to ground
VDD	connected to a +5V power supply
VO	to adjust the contrast
RS	A register select pin that controls where in the LCD's memory you are writing data to. You can select either the data register, which holds what goes on the screen, or an instruction register, which is where the LCD's controller looks for instructions on what to do next.
R/W	A Read/Write pin to select between reading and writing mode
E	An enabling pin that reads the information when High level (1) is received. The instructions are run when the signal changes from High level to Low level.
D0-D7	to read and write data
A	Pins that control the LCD backlight. Connect A to 3.3v.
K	Pins that control the LCD backlight. Connect K to GND.

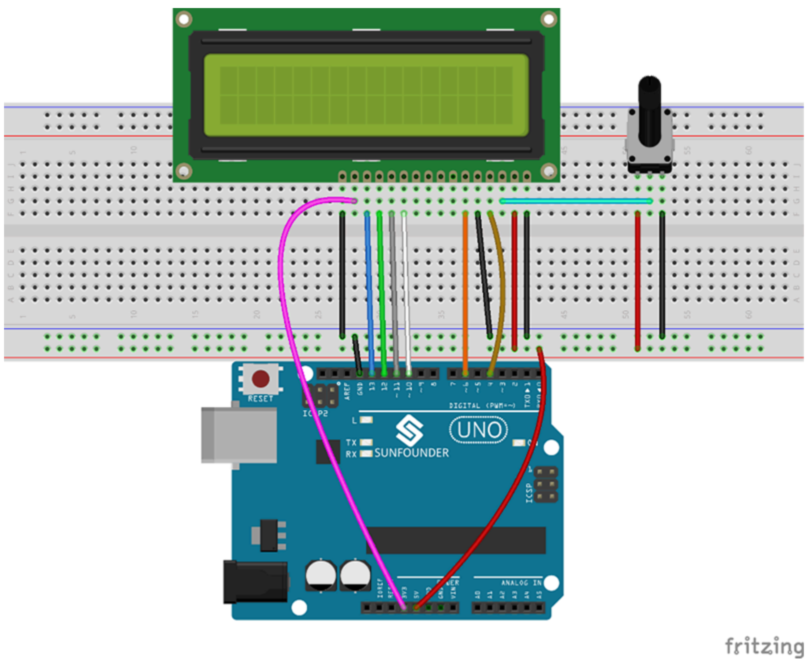
The Experiment for Arduino

Components
- 1 * SunFounder Uno Board
- 1 * Breadboard
- 1 * LCD1602
- 1 * Potentiometer (50kΩ)
- 1 * USB Cable
- Several Jumper Wires

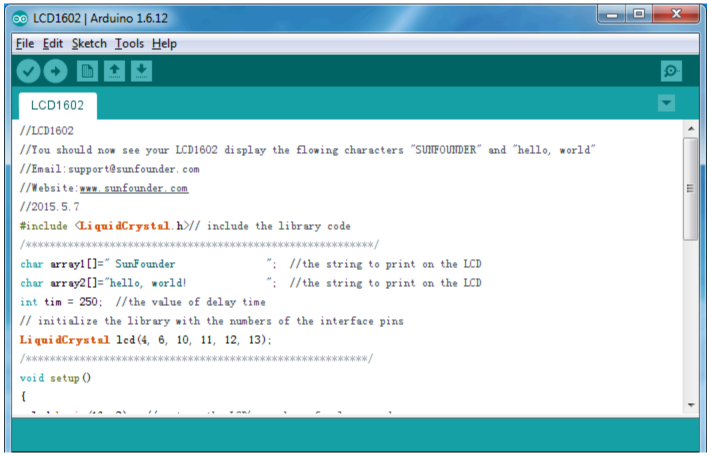
Experimental Procedures
Note: Before connecting circuit, need to plug the pin headers onto a breadboard, and then put the LCD1602 on to it for easy soldering.



Step 1: Build the circuit (make sure the pins are connected correctly. Otherwise, characters will not be displayed properly):

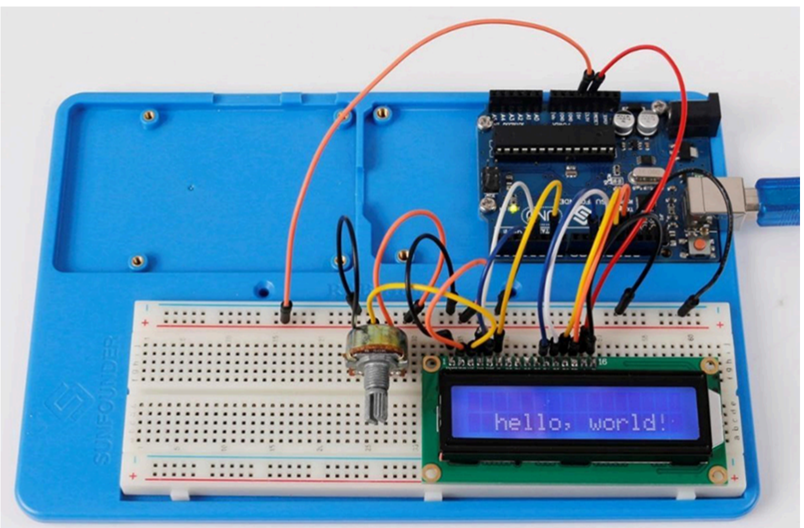


Step 2: Download the package LCD1602_for_Arduino (http://wiki.sunfounder.cc/images/b/b2/LCD1602_for_Arduino.rar), then unzip it and open the LCD1602.ino file



Step 3: Select correct Board and Port
Step 4: Upload the sketch to the SunFounder Uno board

Experimental Phenomenon
Note: You may need to adjust the potentiometer on the LCD1602 until it can display clearly.
You should now see the characters "SunFounder" and "hello, world! " rolling on the LCD.

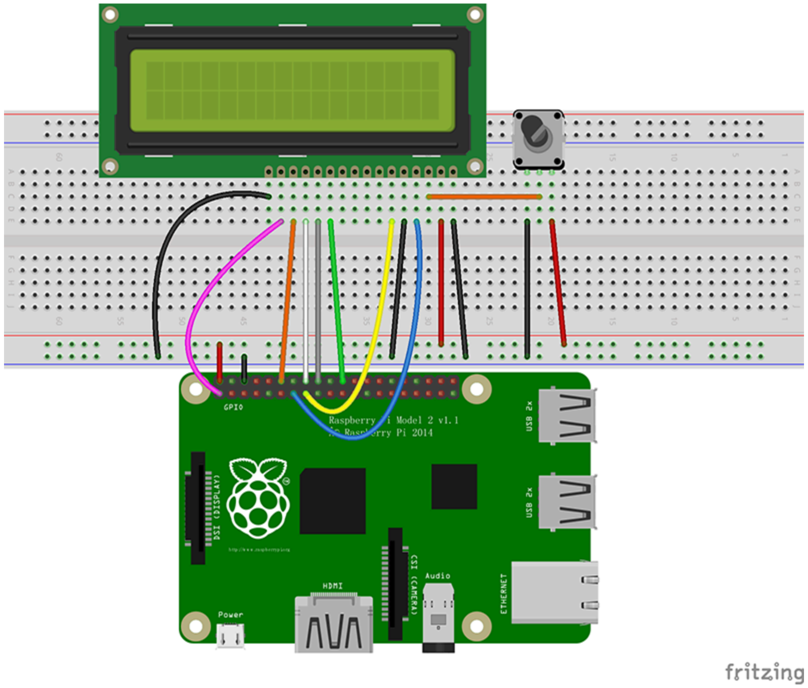


The Experiment for Raspberry Pi

Components
- 1 * Raspberry Pi
- 1 * Breadboard
- 1 * LCD1602
- 1 * Potentiometer
- Several Jumper Wires

Experimental Procedures
Step 1: Build the circuit (please be sure the pins are connected correctly. Otherwise, characters will not be displayed properly):

LCD1602/2004	SunFounder Uno board
K	GND
A	3.3V
D7	GPIO18
D6	GPIO23
D5	GPIO24
D4	GPIO25
D0-D3	No Connected
E	GPIO22
R/W	GND
RS	GPIO27
OV	Connect to the middle pin of potentiometer
VDD	5V
VSS	GND



Note: After you run the code, characters may not appear on the LCD1602. You need to adjust the contrast of the screen (the gradual change from black to white) by spinning the potentiometer clockwise or anticlockwise, until the screen displays characters clearly.

Step 2: Transfer the package LCD1602_for_Raspberry_Pi (http://wiki.sunfounder.cc/images/8/87/LCD1602_for_Raspberry_Pi.zip) to the Raspberry Pi

```
wget http://wiki.sunfounder.cc/images/8/87/LCD1602_for_Raspberry_Pi.zip
```

Step 3: Extract the package

```
unzip LCD1602_for_Raspberry_Pi.zip
```

(For C Language Users)
Step 4: Get into the folder of code

```
cd LCD1602_for_Raspberry_Pi/C
```

Step 5: Compile

```
gcc lcd1602.c -o lcd1602 -lwiringPiDev -lwiringPi
```

Step 6: Run

```
sudo ./lcd1602
```

(For Python Users)

Step 4: Get into the folder of code

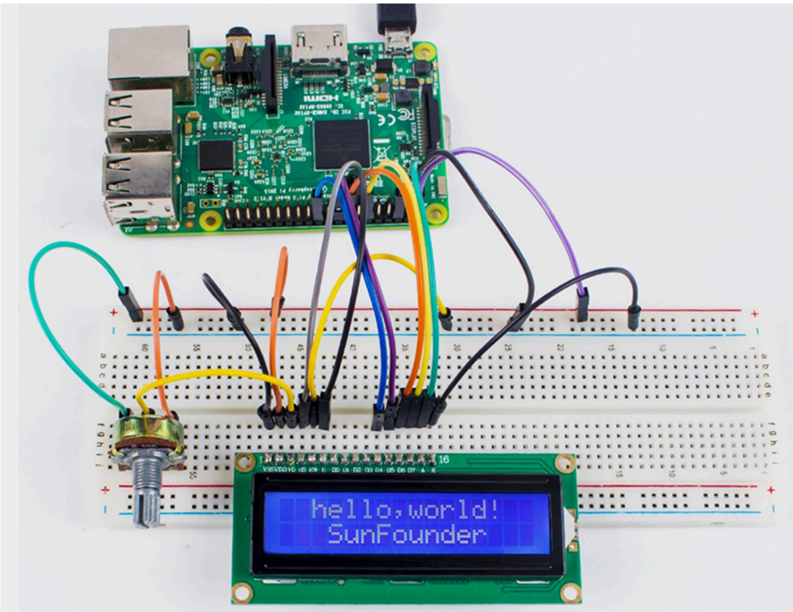
```
cd LCD1602_for_Raspberry_Pi/Python
```

Step 5: Run

```
sudo python lcd1602.py
```

Experimental Phenomenon

You should see two lines of characters displayed on the LCD1602: “hello, world! ” , “SunFounder”.



Resource

LCD1602_for_Arduino (http://wiki.sunfounder.cc/images/b/b2/LCD1602_for_Arduino.rar)

LCD1602_for_Raspberry_Pi (http://wiki.sunfounder.cc/images/8/87/LCD1602_for_Raspberry_Pi.zip)

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